Derivative Obstructions, Minimal Ehrhart Counterexamples, and Peak-Transfer Bounds

Five theorems with exact prior-problem applications

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24 October 2026

Research manuscript T81–T85

Abstract

We give five complete arguments in three settings. For a family of Stern polynomials, a derivative certificate proves that the monotonicity clause of Ulas's Conjecture 4.2(1) fails for every $n \geq 13$ at the fixed parameter $k = 20$; in fact eventual monotonicity fails for every even $k \geq 16$. For equivariant Ehrhart theory, projection fibers certify an unbounded family of minimal counterexamples in dimension ten, addressing Clarke's Question 6.6 with a specific priority qualification. A parity boundary formula then gives a negative answer to Clarke's Question 6.7 and proves effectiveness of every polynomial equivariant numerator for an order-two action in dimension three. For quantum walks, we determine the exact minimum squared-edge-weight cost for nonzero peak transfer at prescribed distance and time. An integral gap-group argument transfers this to adjacency and Laplacian edge bounds for Open Problem 7.2 of Coutinho–Guo–Schmeits. The adjacency bound has optimal cubic leading constant 1/24 and determines the growth order of the minimum. All arguments are unconditional; their prior ingredients and the limits of the historical-novelty assessment are stated explicitly.

1 Exact targets and scope

The workflow in this manuscript is to formulate an auxiliary construction or optimization problem, prove a quantitative statement about it, and apply that statement to a previously published question without changing the question's hypotheses. A negative answer is distinguished from a proof of the proposed assertion. Supporting results are not counted as independently posed conjectures.

Label	Result	Prior target or role
T81	A uniform negative derivative for an infinite terminal range of Stern polynomials.	Refutes the monotonicity clause of Ulas Conjecture 4.2(1) [1].
T82	Minimal ten-dimensional counterexamples with unbounded point count, volume, and h_2^*.	Negative answer and quantitative strengthening for Clarke Question 6.6 [2]; prior-example overlap is explicit.
T83	Polynomiality excludes obstructing faces in dimension three.	Negative answer to Clarke Question 6.7 [2].
T84	Exact weighted energy minimum for arbitrary static graphs with nonzero peak transfer.	Auxiliary optimization used in T85; the attaining chain and its cost are prior.
T85	Adjacency cubic and Laplacian quadratic edge bounds; sharp adjacency leading constant.	Answers the lower-bound request in Open Problem 7.2 [4].

The numbers T81–T85 label five theorems, not five independent formerly open problems. T81 leaves the separate one-real-root assertion untouched. T83 does not determine the least dimension of an ineffective polynomial numerator. T85 gives lower bounds and the adjacency growth order, not the exact minimum edge count for each finite distance.

This is a research draft checked internally by independent derivations, not an externally refereed paper or a formal proof-assistant development. No theorem below assumes an unproved conjecture. Historical priority is a different question: the proofs use classical identities, reciprocity, joins, spectral moments, and Galois arguments. In particular, the bare negative answer to Question 6.6 can also be extracted from a known example once its minimality is supplied. We therefore do not certify that all of the requested historical success conditions have been met, or that a historically unprecedented foundational principle has been discovered.

Ehrhart conventions. For an n-dimensional lattice polytope P at height one, write

$$\sum_{j\geq 0} \#(jP \cap L)t^j = \frac{h_P^*(t)}{(1-t)^{n+1}}.$$

The normalized volume is $h_P^*(1)$, and $\#(P \cap L) = n + 1 + [t]h_P^*(t)$. For an involution σ, let χ_0, χ_1 be the trivial and sign characters, and let ρ be the homogeneous ambient representation. If $E_P(t)$ records the permutation representation on the lattice points of each dilation, its equivariant numerator is

$$\varphi_P(t) = \det(I - t\rho)E_P(t) \in R(\langle\sigma\rangle)[[t]].$$

Its identity evaluation is $h_P^*(t)$. A coefficient is effective precisely when both its trivial and sign multiplicities are nonnegative. If the ambient eigenspace dimensions are p, q, then $\det(I - t\rho)(\sigma) = (1-t)^p(1+t)^q$. These are the conventions in [2].

2 An eventual derivative obstruction for Stern polynomials

The target is Conjecture 4.2(1) of Ulas [1], first circulated in 2019. Write $B_0(t) = 0$, $B_1(t) = 1$, and

$$B_{2m}(t) = tB_m(t), \qquad B_{2m+1}(t) = B_m(t) + B_{m+1}(t).$$

For integers $k \geq 2$, $n \geq 1$, the source defines

$$p_{k,n} = 2^{2n+k} - 3 \cdot 2^{n+k-1} + 2^k - 3, \qquad V_{k,n}(t) = B_{p_{k,n}}(t).$$

The conjecture asserts that for each even k there exists c_k such that, for all $n \geq c_k$, the polynomial $V_{k,n}$ has exactly one real root and is increasing on the entire real line. We disprove its monotonicity clause with a fixed even k and an infinite terminal range of n. A finite counterexample at one n would not suffice to disprove this eventual assertion.

The auxiliary problem is to certify the eventual sign of a derivative when the polynomial degree increases but its rational expression has only a fixed number of powers depending on the index. The following certificate makes the quantifier over all sufficiently large indices explicit.

Lemma 2.1 (A finite derivative certificate)**.** *Let H_0, H_1, H_2 be real differentiable functions near a fixed real number a with $0 < r := |a| < 1$. Suppose*

$$F_n(t) = H_0(t) + H_1(t)t^{n-1} + H_2(t)t^{2n-2}.$$

Then

$$|F_n'(a) - H_0'(a)| \le \left(|H_1'(a)| + \frac{(n-1)|H_1(a)|}{r}\right) r^{n-1} + \left(|H_2'(a)| + \frac{2(n-1)|H_2(a)|}{r}\right) r^{2n-2}.$$

In particular, $F_n'(a) \longrightarrow H_0'(a)$. If $H_0'(a) < 0$, then F_n is not increasing on the real line for all sufficiently large n.

Proof. Differentiate the displayed identity and apply the triangle inequality. Since $0 < r < 1$, both nr^{n-1} and nr^{2n-2} tend to zero. A differentiable increasing real function has nonnegative derivative wherever its derivative exists, which proves the last assertion. □

Theorem T81 (Failure of Ulas's eventual monotonicity conjecture)**.** *For every integer $n \ge 13$,*

$$B'_{2^{2n+20}-3\cdot 2^{n+19}+2^{20}-3}(-3/4) < 0.$$

Consequently no constant c_{20} has the monotonicity property required by Conjecture 4.2(1) of Ulas, and that original conjecture is false.

Proof. We first record the exact polynomial expression needed for the certificate. The Stern addition identity is

$$B_{m2^a \pm r}(t) = B_{2^a - r}(t)B_m(t) + B_r(t)B_{m\pm 1}(t) \qquad (0 \le r \le 2^a), \tag{1}$$

with the minus version used only when its indices are nonnegative. It follows by induction from the defining recurrences and is also [1, Lemma 2.1]. The geometric-sum identities give

$$B_{2^k-3}(t) = \frac{2t^{k-1} - t - 1}{t-1}, \quad B_{2^{n+1}-3}(t) = \frac{2t^n - t - 1}{t-1}, \quad B_{2^{n+1}-2}(t) = \frac{t(t^n-1)}{t-1}.$$

All quotients represent polynomials; we will evaluate only at $t = -3/4$. Writing

$$p_{k,n} = 2^{n+k-1}(2^{n+1} - 3) + (2^k - 3)$$

and then using (1) on the complementary index, we obtain

$$V_{k,n}(t) = \left((t+1)t^{n-1} + \frac{(2t^{k-1} - t - 1)(t^{n-1} - 1)}{(t-1)^2}\right) \frac{2t^n - t - 1}{t-1} + \frac{t(2t^{k-1} - t - 1)(t^n - 1)}{(t-1)^2}. \tag{2}$$

This expression already occurs in the proof of [1, Theorem 2.3]; the derivative obstruction below is the additional step.

Put $u = t^{n-1}$. Direct collection of its three powers gives

$$(t-1)^3 V_{k,n}(t) = C_0(t) + C_1(t)u + C_2(t)u^2, \tag{3}$$

where

$$\begin{aligned} C_0(t) &= (2t^{k-1} - t - 1)(-t^2 + 2t + 1), \\ C_1(t) &= -2t^4 + 6t^2 + 4t + (2t^3 - 2t^2 - 6t - 2)t^{k-1}, \\ C_2(t) &= 2t^4 - 2t^3 - 4t^2 + 4t^k. \end{aligned}$$

Fix $k = 20$, let $H_j = C_j/(t-1)^3$, and set $a = -3/4$, $r = 3/4$. Exact rational substitution gives

$$H_0'(a) = -\frac{110236939137}{1473173782528} < -\frac{7}{100}.$$

For the other quantities in Lemma 2.1, the exact values and simple upper bounds are

$$|H_1'(a)| = \frac{4432120404109}{11785390260224} < \frac{1}{2}, \qquad \frac{|H_1(a)|}{r} = \frac{108928627487}{1683627180032} < \frac{1}{10},$$
$$|H_2'(a)| = \frac{164405164071}{368293445632} < \frac{1}{2}, \qquad \frac{2|H_2(a)|}{r} = \frac{9957814131}{26306674688} < \frac{1}{2}.$$

Thus

$$|V_{20,n}'(a) - H_0'(a)| < E_n := \frac{n+4}{10}\left(\frac{3}{4}\right)^{n-1} + \frac{n}{2}\left(\frac{3}{4}\right)^{2n-2}.$$

Both summands decrease for $n \geq 2$: their consecutive ratios are, respectively,

$$\frac{3}{4}\frac{n+5}{n+4} < 1, \qquad \frac{9}{16}\frac{n+1}{n} < 1.$$

Finally, the rational inequality

$$E_{13} = \frac{17}{10}\left(\frac{3}{4}\right)^{12} + \frac{13}{2}\left(\frac{3}{4}\right)^{24} < \frac{7}{100}$$

shows that $V_{20,n}'(a) < 0$ for every $n \geq 13$. This excludes every possible choice of c_{20} and proves the theorem. □

The mechanism can also be seen in the limiting profile on $0 < x < 1$:

$$\lim_{n\to\infty} V_{k,n}(-x) = \frac{(2x^{k-1} - x + 1)(1 - 2x - x^2)}{(1+x)^3} \quad \text{for even } k.$$

Its derivative at $x = 3/4$ is positive for $k = 20$, which is incompatible with eventual increase in $t = -x$. The finite certificate above upgrades this observation to the uniform threshold $n \geq 13$.

In fact the failure occurs for every even $k \geq 16$. The derivative of the limiting profile at $x = 3/4$ is exactly

$$\frac{8}{2401}\left(36 - (119k + 22)\left(\frac{3}{4}\right)^{k-2}\right).$$

At $k = 16$ it equals $32262723/5754585088 > 0$, and the sequence $(119k+22)(3/4)^{k-2}$ decreases for integer $k \geq 3$. Thus, for each even $k \geq 16$, Lemma 2.1 gives a terminal range of n on which $V_{k,n}'(-3/4) < 0$. Only the explicit threshold $n \geq 13$ for $k = 20$ is needed for the theorem.

The result refutes the entire conjunction in the stated conjecture through one of its required clauses. It does not assert that the separate single-real-root clause is false. The proof uses a prior Stern identity, rational differentiation, and a geometric error estimate. These tools are classical; this argument is not presented as a historically unprecedented foundational principle.

3 Projection fibers and unbounded minimal counterexamples

Clarke's Question 6.6 [2] asks whether a minimal counterexample to effectiveness must have $n+2$ lattice points, normalized volume four, and zero coefficient of t^2 in its ordinary h^*-polynomial. Here minimality means that no proper full-dimensional invariant lattice subpolytope with the

same fixed polytope remains. We answer this exact question negatively by a family in fixed dimension ten, with all three quantities unbounded.

The auxiliary problem is to certify such minimality from the fibers of the averaging projection. This avoids enumerating invariant subpolytopes, which may contain nonvertex lattice points. The underlying free-join and pole-cancellation construction is prior: Ziegler's examples, recalled in [2, Example 9.2], already multiply the reflected-square series by an ordinary h^*-polynomial. The contribution proposed here is the minimality certificate and the following explicit unbounded family.

Lemma 3.1 (Projection-fiber forcing)**.** *Let P be an invariant lattice polytope with involution σ, and let $\pi = (I+\sigma)/2$. Suppose that for each vertex w of P^σ, the lattice points of $P \cap \pi^{-1}(w)$ constitute a single σ-orbit O_w, and that these orbits together contain every vertex of P. Then no proper invariant lattice subpolytope P' has $(P')^\sigma = P^\sigma$.*

Proof. Since P' is convex and invariant, $\pi(P') = (P')^\sigma$. A vertex w of this image must be the image of at least one vertex of P'. That vertex is a lattice point in O_w, and invariance of P' forces the entire orbit O_w into P'. Consequently P' contains every vertex of P, and $P' = P$. □

For $m \geq 1$, define the five-dimensional lattice simplex

$$Q_m = \operatorname{conv}\{0, e_1, e_2, e_3, e_4, (1, -1, 1, -m, 2m)\} \subset \mathbb{R}^5. \tag{4}$$

Let $S = [0,1]^2$ with involution $(x, y) \mapsto (1 - x, y)$, and let $I_m = [0, m+2]$ with trivial action. Use the standard lattice free join: take the direct sum of the homogeneous lattices, use the sum of their height functionals as height, and take the convex hull of the embedded height-one polytopes. Put

$$R_m = S * Q_m * I_m,$$

with the involution trivial on Q_m and I_m.

Theorem T82 (An unbounded negative answer to Clarke's Question 6.6)**.** *For every $m \geq 1$, the polytope R_m has dimension ten and is minimal in the exact sense of [2, Question 6.6]. Its equivariant numerator is polynomial but not effective. More precisely, putting*

$$q_m(t) = 1 + mt - t^2 + m(m+1)t^3,$$

its trivial and sign characters χ_0, χ_1 occur with the virtual multiplicities

$$\varphi_{R_m}(t) = (1 + t + t^2)q_m(t)\chi_0 + tq_m(t)\chi_1. \tag{5}$$

The sign multiplicity at degree three equals -1, whereas

$$\#(R_m \cap L) = m + 13, \qquad \operatorname{Vol}_{\mathrm{norm}}(R_m) = 4m(m+2), \qquad [t^2]h^*_{R_m}(t) = 2m. \tag{6}$$

Thus none of the three proposed necessary properties in Question 6.6 holds, and their failure is unbounded in fixed dimension.

Proof. The homogeneous vertex matrix of Q_m has determinant of absolute value $2m$. The barycentric vector

$$\frac{1}{2m}(m, -1, 1, -1, m, 1)$$

maps to the integral homogeneous point $(e_5, 1)$, with the six coordinates ordered as in (4). Its last coordinate has exact denominator $2m$, so it generates the quotient of order $2m$. The

fundamental parallelepiped lattice points are therefore represented by the fractional parts of its multiples k, $0 \le k < 2m$.

At $k = 0$, the resulting height is zero. For nonzero even k, each of the two pairs with opposite weights sums to one, while the two m-weighted coordinates vanish. The height is two, for $m - 1$ values of k. For odd k, the latter two coordinates sum to one, so the height is three, for m values of k. The fundamental parallelepiped formula gives

$$h^*_{Q_m}(t) = 1 + (m-1)t^2 + mt^3 = (1+t)(1-t+mt^2). \tag{7}$$

In particular Q_m has exactly its six vertices as lattice points.

The reflected square has

$$\varphi_S(t) = \frac{1+t+t^2}{1+t}\chi_0 + \frac{t}{1+t}\chi_1,$$

as also recorded in [2, Examples 2.3 and 2.8]. The interval has ordinary numerator $1 + (m + 1)t$. Under the standard lattice join, cone semigroups are direct products and the ambient representations are direct sums. Thus Hilbert series and determinant factors multiply, and so do equivariant numerators. Applying (7) cancels the square denominator and gives (5), since

$$(1-t+mt^2)(1+(m+1)t) = 1 + mt - t^2 + m(m+1)t^3.$$

The degree-three sign multiplicity is -1, proving failure of effectiveness despite polynomiality.

The dimension of the three-factor free join is $2 + 5 + 1 + 2 = 10$. Evaluating its numerator at the identity gives

$$h^*_{R_m}(t) = (1+t)^2 q_m(t).$$

Since $[t]h^*_{R_m} = m+2$, the standard degree-one Ehrhart identity gives $\#(R_m \cap L) = 11+m+2 = m + 13$. Its degree-two coefficient is $2m$, and its value at 1 is $4m(m + 2)$. These prove (6).

It remains to check minimality, including possible subpolytopes that use nonvertex lattice points. The fixed polytope is the free join $S^\sigma * Q_m * I_m$. Each vertex belonging to Q_m or I_m has a singleton fiber under π: its square-factor homogeneous height is zero, and the involution is trivial on the other factors. The two vertices of the fixed square segment are $(1/2, 0)$ and $(1/2, 1)$. Their fibers are the two unit horizontal edges of S. Each contains exactly its two lattice endpoints, and those endpoints form a single σ-orbit. The direct-sum height coordinates prevent any other join factor from contributing to these fibers. These fibers account for all vertices of R_m. Lemma 3.1 now gives minimality, even without imposing the full-dimensionality restriction on competing subpolytopes. □

For example, $m = 1$ gives a ten-dimensional polytope with 14 lattice points, normalized volume 12, and $h^*_2 = 2$. Its numerator is

$$(1 + 2t + t^2 + 2t^3 + t^4 + 2t^5)\chi_0 + (t + t^2 - t^3 + 2t^4)\chi_1.$$

The exact verification bundle independently enumerates the fundamental parallelepiped and counts lattice points in Q_m and $2Q_m$ for $1 \le m \le 12$. Those counts are 6 and $m + 20$, respectively.

Priority boundary. The known thirteen-dimensional example in [2, Example 9.2] also admits the same fiber-minimality argument. Thus the bare negative answer can already be extracted from that example once its minimality is supplied. The present result provides an explicit fixed-dimensional unbounded family and a full minimality proof; it does not claim that the reflected-square join mechanism or all its counterexample inputs are new. This limits any claim to a historically new methodology, even though the original yes/no question is answered.

4 Half-lattice boundary detection and Clarke's dimension-three question

Clarke's Question 6.7 [2] asks whether a three-dimensional lattice polytope with an involution of integral type $(2, 0, 1)$ can have a polynomial equivariant numerator while possessing an obstructing face. We give a negative answer to this exact question. Our auxiliary problem is to detect, by a boundary coefficient, an edge of a half-lattice polygon which has no lattice point. This detection prevents the period collapse that would be required by the proposed example.

The noncollapse of the second Ehrhart coefficient is established in the literature; see Beck–Sam–Woods [3]. We give the special parity formula needed here directly. The additional geometric step connects the source's nonsaturated face-generated groups to that formula without replacing them by saturated lattices.

4.1 A segment lemma and an exact parity formula

For a lattice Λ, write $\frac{1}{2}\Lambda$ for its half-lattice. The same notation on an affine lattice means that an affine lattice origin has been chosen.

Lemma 4.1 (Half-lattice segment detection)**.** *Let $E = [u, v]$ be a nondegenerate segment with $u, v \in \frac{1}{2}\Lambda$. If its affine line contains a point of Λ, then E contains a point of Λ.*

Proof. Choose $x_0 \in \Lambda \cap \operatorname{aff}(E)$ and a primitive lattice vector w parallel to E. Since $2(u - x_0), 2(v - x_0) \in \Lambda \cap \mathbb{R}w = \mathbb{Z}w$, there are distinct integers a, b with $u = x_0 + (a/2)w$ and $v = x_0 + (b/2)w$. A closed interval with distinct half-integer endpoints contains an integer. If the endpoint parities differ, one endpoint is integral; if both endpoints are odd halves, their distance is at least one. The resulting integral parameter gives a lattice point of E. □

Let R be a two-dimensional polygon with half-lattice vertices in an affine rank-two lattice Λ. Its Ehrhart constituents have the form

$$N_{\mathrm{even}}(k) = Ak^2 + B_{\mathrm{even}}k + C_{\mathrm{even}}, \qquad N_{\mathrm{odd}}(k) = Ak^2 + B_{\mathrm{odd}}k + C_{\mathrm{odd}}.$$

For an edge E, let $\lambda(E) > 0$ denote its length measured in the primitive lattice direction parallel to it. Write its supporting line as $\langle n_E, x\rangle = c_E$, where n_E is primitive integral. The half-lattice vertices imply $2c_E \in \mathbb{Z}$.

Lemma 4.2 (Parity boundary formula)**.** *With this notation,*

$$B_{\mathrm{even}} - B_{\mathrm{odd}} = \frac{1}{2} \sum_{E:\, c_E \notin \mathbb{Z}} \lambda(E). \tag{8}$$

In particular, a non-lattice supporting line forces a strictly positive difference between the linear Ehrhart coefficients.

Proof. Rational Ehrhart reciprocity in dimension two gives $\#(\operatorname{relint}(kR) \cap \Lambda) = N(-k)$. Since k and $-k$ have the same parity, subtraction shows that the number of boundary lattice points is exactly $2B_{\mathrm{even}}k$ for even positive k, and $2B_{\mathrm{odd}}k$ for odd positive k.

For even k, every line $\langle n_E, x\rangle = kc_E$ contains lattice points, and the number on kE is $k\lambda(E) + O(1)$. For odd k, this remains true when $c_E \in \mathbb{Z}$; when $c_E \notin \mathbb{Z}$, the supporting line

contains no lattice point. Summing over edges overcounts vertices by a bounded amount. Comparing linear terms in the two parity classes yields

$$2B_{\text{even}} = \sum_{E} \lambda(E), \qquad 2B_{\text{odd}} = \sum_{E:\, c_E \in \mathbb{Z}} \lambda(E).$$

Their difference proves (8). □

4.2 The unchanged prior question

Use Clarke's homogeneous convention: L is a rank-four lattice with a primitive height functional, P is a three-dimensional lattice polytope at height one, and σ is a height-preserving involution. Type $(2, 0, 1)$ means that its integral normal form has two trivial one-dimensional blocks and one block exchanging two coordinates. The ambient eigenspace dimensions are therefore $p = 3$ and $q = 1$.

For an invariant face F, let $M_F = \mathbb{Z}\langle F \cap L\rangle$. The face is *$\sigma$-odd* when every element of M_F^σ has even height, and is *obstructing* when it is σ-odd and $\dim F^\sigma \geq q$. These are the original definitions [2, Definition 4.1]; $F = P$ is permitted and M_F need not be saturated. Write $\varphi_P(t)$ for the equivariant numerator.

Theorem T83 (Negative answer to Clarke's Question 6.7)**.** *For every three-dimensional lattice polytope with an involution of type $(2, 0, 1)$, polynomiality of $\varphi_P(t)$ implies that P has no obstructing face. Consequently no polytope requested in [2, Question 6.7] exists.*

More generally, for every three-dimensional lattice polytope with an order-two action, polynomiality of its equivariant numerator implies the existence of an invariant linear system of parameters of the prescribed ambient representation type, and hence effectiveness of that numerator.

Proof. Let $R = P^\sigma$. Its dimension is two and its vertices are orbit averages $(v + \sigma(v))/2$, by [2, Lemma 2.2]. The fixed lattice L^σ has rank three, and its height group is either $\mathbb{Z}$ or $2\mathbb{Z}$.

If this height group is $2\mathbb{Z}$, the fixed lattice-point count is zero at every odd dilation. At the even dilations it is a degree-two polynomial with positive leading coefficient. The fixed Ehrhart series therefore has a genuine pole of order three at $t = -1$. The character identity

$$\varphi_P(t)(\sigma) = (1 - t)^3(1 + t) \sum_{k \geq 0} \#(kR \cap L)\, t^k \tag{9}$$

cancels only one order of this pole, contradicting polynomiality.

We may thus suppose that the height group is $\mathbb{Z}$. Choose a fixed height-one lattice point as affine origin; it need not lie in P. This identifies the fixed height-one affine lattice with a rank-two lattice Λ, and the orbit-average formula shows that R has half-lattice vertices.

Suppose that F is an obstructing face. The face contains no σ-fixed ambient lattice point: such a point would itself belong to $F \cap L \subset M_F$ and have height one, contrary to σ-oddness. This implication makes no saturation assumption on M_F.

If $F \neq P$, then F^σ is a proper face of R by [2, Lemma 2.2(b)]. Its dimension is at least $q = 1$, so it is an edge. It contains no lattice point. Lemma 4.1 implies that its supporting affine line also contains no lattice point, and Lemma 4.2 gives $B_{\text{even}} - B_{\text{odd}} > 0$. If $F = P$, then R contains no lattice point at all, so the same argument applies to each of its edges and gives the same strict inequality. This also handles a nonsaturated group generated by all lattice points of P.

The leading area coefficients of the two Ehrhart constituents agree. Their difference is thus a degree-one polynomial with positive linear coefficient. Write the fixed count as

$$N(k) = \frac{N_{\mathrm{even}}(k) + N_{\mathrm{odd}}(k)}{2} + (-1)^k \frac{N_{\mathrm{even}}(k) - N_{\mathrm{odd}}(k)}{2}.$$

The generating function of the first term has poles only at $t = 1$. The second term has a genuine pole of order two at $t = -1$, since its linear coefficient is nonzero. Formula (9) cancels just one order, so $\varphi_P(t)(\sigma)$ is not a polynomial. This contradiction rules out every obstructing face and proves the exact negative answer.

By [2, Theorem 4.2], absence of obstructing faces gives the claimed linear system of parameters and effectiveness. For the extension to all order-two actions in dimension three, if $p \le q$ the same conclusion is [2, Corollary 4.4]. The remaining possibility is $(p, q) = (3, 1)$. Besides type $(2, 0, 1)$, its only integral type is $(3, 1, 0)$; in that type the fixed polytope is integral, and [2, Corollary 4.5] applies. These cases exhaust the possible nontrivial actions. The trivial action is the ordinary Ehrhart case. □

The final dimension-three statement is a consequence of the one new prior-problem application, not another independently posed question. It raises the order-two lower bound in Clarke's Question 6.8 from three to four, but does not determine the least counterexample dimension: dimensions four and five are not settled here.

The finite verification accompanying this argument uses exact integer and rational arithmetic on 52 half-lattice polygons. It checks both Ehrhart constituents and (8), including a polygon with period collapse at a nonintegral vertex. These checks support the implementation of the formula; the preceding lemmas provide its proof for every polygon.

5 An exact energy problem for peak state transfer

Coutinho, Guo, and Schmeits ask in Open Problem 7.2 of [4] for a lower bound on the number of edges of a graph admitting peak state transfer between vertices at distance d. The statement does not specify adjacency or Laplacian dynamics, so both are treated below. Our auxiliary problem first permits real edge weights: at a fixed transfer time and distance, what is the least possible sum of squared edge weights? It has an exact answer. A Galois argument then removes the time parameter for the original unweighted problem.

The ingredients have established antecedents. Coutinho's size bounds [5] use spectral spacing and trace moments for periodic vertices. The arithmetic characterization of mutual spectral support is in [4, Theorem 5.4]. The chain attaining our weighted bound is the classical hypercube quotient underlying engineered perfect transfer [6]; its fixed-energy transfer time, including the same constant, is explicitly recorded in [7, equation (17)]. The proposed contribution is the exact optimization for arbitrary nonzero peak transfer, followed by the bounds and sharpness for the original sparsity question; these antecedents are not presented as a new foundational principle.

For a real symmetric matrix M, let $M = \sum_\theta \theta E_\theta$. For distinct indices u, v, put

$$S = S_M(u, v) = \{\theta : (E_\theta)_{uv} \neq 0\}, \qquad c_\theta = (E_\theta)_{uv}, \qquad B_M(u, v) = \sum_{\theta \in S} |c_\theta|.$$

Nonzero peak transfer at time $\tau > 0$ means

$$\left|(e^{i\tau M})_{uv}\right| = B_M(u, v) > 0.$$

Equality in the triangle inequality is equivalent to the existence of a unit complex number γ such that

$$e^{i\tau\theta} = \gamma \operatorname{sgn}(c_\theta) \qquad (\theta \in S). \tag{10}$$

In particular, any two supported eigenvalues differ by an integer multiple of π/τ.

Lemma 5.1 (Distance and mutual support)**.** *Suppose the graph of nonzero off-diagonal entries of M has distance d between u and v, and $B_M(u,v) > 0$. Then $|S_M(u,v)| \geq d+1$.*

Proof. For $0 \leq j < d$, a product contributing to $(M^j)_{uv}$ cannot traverse the required d off-diagonal steps, even if diagonal entries are allowed. Thus

$$0 = (M^j)_{uv} = \sum_{\theta\in S} c_\theta \theta^j \qquad (0 \leq j < d).$$

If $s = |S| \leq d$, the first s equations form a square Vandermonde system on the distinct elements of S. Its invertibility would force all c_θ to be zero, a contradiction. □

Lemma 5.2 (Quadratic cost of separated points)**.** *If $x_1 < \cdots < x_s$ and $x_{j+1} - x_j \geq a > 0$, then*

$$\sum_{j=1}^{s} x_j^2 \geq \frac{a^2 s(s^2-1)}{12}. \tag{11}$$

Equality holds precisely for $x_j = a(j - (s+1)/2)$.

Proof. Writing $\bar{x} = s^{-1}\sum_j x_j$ gives

$$\sum_j x_j^2 = s\bar{x}^2 + \frac{1}{s}\sum_{i<j}(x_j - x_i)^2 \geq \frac{a^2}{s}\sum_{i<j}(j-i)^2 = \frac{a^2 s(s^2-1)}{12}.$$

Equality requires $\bar{x} = 0$ and all consecutive gaps equal to a, which is also sufficient. □

Theorem T84 (Exact fixed-time energy minimum)**.** *Let G be a finite loopless graph with nonzero real edge weights w_e that are constant in time, and let A be its weighted adjacency matrix. If nonzero peak transfer occurs across distance $d \geq 1$ at time $\tau > 0$, then*

$$\tau^2 \sum_{e\in E(G)} w_e^2 \geq \frac{\pi^2 d(d+1)(d+2)}{24}. \tag{12}$$

For every $d \geq 1$ and $\tau > 0$, equality is attained by a weighted path on $d+1$ vertices with consecutive weights

$$w_j = \frac{\pi}{2\tau}\sqrt{j(d+1-j)}, \qquad 1 \leq j \leq d. \tag{13}$$

Thus the right side is the exact optimum over all such weighted graphs, with no assumed lower bound on the peak amplitude other than positivity.

Proof. Put $s = |S_A(u,v)|$. By Lemma 5.1, $s \geq d+1$; by (10), its ordered eigenvalues have gaps at least π/τ. Each supported eigenvalue occurs in the full spectrum with multiplicity at least one. Therefore

$$2\sum_e w_e^2 = \operatorname{tr}(A^2) \geq \sum_{\theta\in S}\theta^2 \geq \frac{\pi^2 s(s^2-1)}{12\tau^2} \geq \frac{\pi^2 d(d+1)(d+2)}{12\tau^2}.$$

This proves the lower bound.

For attainment, put $a = \pi/\tau$ and consider on $(\mathbb{C}^2)^{\otimes d}$ the operator $(a/2)\sum_{j=1}^{d} X_j$, where X_j exchanges the two basis vectors in tensor coordinate j. The subspace of vectors constant on each Hamming-weight class is invariant. In the normalized class basis, its restriction is the path with weights (13). At time τ each tensor factor evolves as $e^{i\pi X/2} = iX$, so the all-zero vector is sent to i^d times the all-one vector. These two vectors are the endpoint class vectors. Consequently the weighted path has perfect, hence peak, transfer at τ. Finally,

$$\sum_{j=1}^{d} w_j^2 = \frac{\pi^2}{4\tau^2} \sum_{j=1}^{d} j(d+1-j) = \frac{\pi^2 d(d+1)(d+2)}{24\tau^2}.$$

The construction and its energy value are established. The lower bound here proves optimality within the larger class of arbitrary static graphs and nonzero peak transfer. It does not apply to time-dependent controls, which can be faster [7]. □

6 Edge bounds for the original peak-transfer problem

The previous optimization still contains the time. For an integer matrix, the mutual support supplies an arithmetic normalization even when neither vertex is periodic. This distinction is essential: peak transfer does not in general imply periodicity at either endpoint.

Lemma 6.1 (The integral gap group)**.** *Let M be a symmetric integer matrix and suppose nonzero peak transfer occurs between $u \neq v$. The additive subgroup generated by the differences $\theta - \theta'$ with $\theta, \theta' \in S_M(u,v)$ is $\beta\mathbb{Z}$, where $\beta > 0$ and β^2 is a positive integer. The positive peak times are precisely the odd multiples of π/β. If additionally $S_M(u,v) \subseteq [0,\infty)$ and $0 \in S_M(u,v)$, then β is a positive integer and all elements of $S_M(u,v)$ are nonnegative integer multiples of β.*

Proof. Let $K \subset \mathbb{R}$ be the splitting field of the characteristic polynomial. Its roots are real algebraic integers. The spectral-projector formula

$$E_\theta = \prod_{\lambda \neq \theta} \frac{M - \lambda I}{\theta - \lambda}$$

shows that every Galois automorphism of $K/\mathbb{Q}$ sends E_θ to $E_{\sigma\theta}$. Hence $S_M(u,v)$ is Galois-stable.

At an existing peak time τ, all differences belong to $(\pi/\tau)\mathbb{Z}$. The group they generate is consequently $\beta\mathbb{Z}$ for some $\beta > 0$, and its positive generator is an integer linear combination of those differences. Thus β is an algebraic integer. Galois stability of S makes this group invariant; its only two generators are β and $-\beta$. Therefore $\sigma\beta = \pm\beta$ for every σ, and β^2 is rational and an algebraic integer. It is a positive integer.

Fix $\theta_0 \in S$ and write $\theta = \theta_0 + n_\theta\beta$. The integers n_θ have greatest common divisor one. The existing peak alignment implies $\tau\beta = k\pi$ for an integer k. Since $\sum_{\theta\in S} c_\theta = (I)_{uv} = 0$ and not all c_θ vanish, both signs occur. Thus k is odd: an even k would give the same phase at every supported eigenvalue. At π/β the phase ratios $(-1)^{n_\theta}$ are the same as at τ, proving peak transfer there. The same reasoning proves exactly the asserted set of peak times.

For the last claim, $0 \in S$ gives $S \subseteq \beta\mathbb{Z}$. If some Galois automorphism sent β to $-\beta$, it would send a positive element of S to a negative element, contradicting $S \subseteq [0,\infty)$. Such a positive element exists because $u \neq v$ and $|S| \geq 2$. Therefore β is fixed by Galois, and is a positive rational algebraic integer, hence a positive integer. □

Theorem T85 (Peak-transfer sparsity bounds and cubic optimality)**.** *Let a finite simple graph have m edges and nonzero peak transfer between vertices at graph distance $d \geq 1$.*

1. *For adjacency dynamics,*
$$m \geq \frac{d(d+1)(d+2)}{24}. \tag{14}$$
The leading constant $1/24$ is optimal: an infinite sequence of unweighted graphs and vertex pairs has $d \to \infty$ and $m/d^3 \to 1/24$.

2. *For Laplacian dynamics,*
$$m \geq \frac{d(d+1)}{4}. \tag{15}$$
No sharpness claim for the Laplacian bound is made here.

In particular these are unconditional lower bounds answering the formal request of [4, Open Problem 7.2]. If $m_A(d)$ denotes the minimum number of edges for adjacency peak transfer across distance d, then $m_A(d) = \Theta(d^3)$. The exact finite-distance minimum is not determined.

Proof. One may restrict to the connected component containing the two vertices; its edge count is no larger than m. For adjacency dynamics, Lemma 6.1 gives a peak time $\pi/\beta \leq \pi$. Theorem T84, with all weights one, gives the stronger bound

$$m \geq \frac{\beta^2 d(d+1)(d+2)}{24} \geq \frac{d(d+1)(d+2)}{24}.$$

For Laplacian dynamics, L is positive semidefinite and its zero projector is J/N on a connected graph of order N. Thus 0 is in mutual support. By Lemma 6.1, the supported eigenvalues are distinct nonnegative integers. If their number is s, then $s \geq d+1$ by Lemma 5.1, which permits the diagonal entries of L. Consequently

$$2m = \operatorname{tr} L \geq \sum_{\theta \in S_L(u,v)} \theta \geq \frac{s(s-1)}{2} \geq \frac{d(d+1)}{2}.$$

More precisely the first gap generator improves this to $m \geq \beta d(d+1)/4$.

We give the sharpness construction and the all-distance upper bound in the following paragraphs, completing the theorem. □

6.1 An unweighted sharpness family, verified at individual vertices

The graphs Y_n in [4, Section 6.3] provide the required upper examples. For completeness, their spectral and transfer properties needed here are verified directly. For $n \geq 2$, take $2n$ independent vertex classes in a chain, with sizes

$$s_{2j-1} = n - j + 1, \qquad s_{2j} = j \qquad (1 \leq j \leq n),$$

and join consecutive classes by complete bipartite graphs, with no other edges. The vertex and edge counts are

$$|V(Y_n)| = n(n+1), \qquad |E(Y_n)| = \sum_{j=1}^{n} j(n-j+1) + \sum_{j=1}^{n-1} j(n-j) = \frac{n(n+1)(2n+1)}{6}. \tag{16}$$

Let x be the unique vertex in class 2, and let y be any one vertex in class $2n$. Their distance is $2n-2$.

Let N have the normalized class indicators as columns. Then $N^TN = I$, and the full adjacency matrix is $A = NQN^T$, where Q is the weighted path with off-diagonal entries $\sqrt{s_j s_{j+1}}$. We first verify that its eigenvalues are $\pm 1, \dots, \pm n$. In the odd/even class ordering, Q has form $\left(\begin{smallmatrix} 0 & C \\ C^T & 0 \end{smallmatrix}\right)$, where

$$C_{jj} = \sqrt{j(n-j+1)}, \qquad C_{j+1,j} = \sqrt{j(n-j)}.$$

The even-coordinate block C^TC has diagonal $j(2n+1-2j)$ and $(j, j+1)$ entry $(n-j)\sqrt{j(j+1)}$. Conjugating by the diagonal matrix with entries $(-1)^j\sqrt{j}$ changes it to $I+\mathcal{L}$, where on functions of $j \in \{1, \dots, n\}$,

$$(\mathcal{L}f)(j) = (n-j)(j+1)\big(f(j) - f(j+1)\big) + (n-j+1)(j-1)\big(f(j) - f(j-1)\big).$$

Boundary terms have zero coefficient. On the polynomial basis $1, j, \dots, j^{n-1}$, $\mathcal{L}$ preserves degree and sends j^k to $k(k+2)j^k$ plus lower-degree terms. Therefore $I + \mathcal{L}$ has eigenvalues $(k+1)^2$, $0 \le k < n$. This proves the claimed spectrum of Q and gives

$$p(z) := \det(zI - Q) = \prod_{k=1}^{n} (z^2 - k^2).$$

The $(2, 2n)$ entry of $(zI - Q)^{-1}$ is

$$\frac{z\sqrt{n}\,((n-1)!)^2}{p(z)}.$$

Indeed, the path cofactor is the product of the weights from positions 2 through $2n$, times the characteristic polynomial z of the single remaining first vertex. Taking residues at $\pm k$ yields

$$(E^Q_{\pm k})_{2,2n} = \frac{\sqrt{n}\,((n-1)!)^2}{2\prod_{1\le j\le n,\, j\ne k}(k^2 - j^2)}.$$

The corresponding projector entry in the full graph is divided by $\sqrt{n}$, since the two endpoint classes have sizes 1 and n. The additional zero projector is $I - NN^T$ and has (x, y) entry zero. Thus every nonzero mutual coefficient has sign $(-1)^{n-k}$, the same at k and $-k$. At time π, multiplication by $e^{\pm ik\pi} = (-1)^k$ gives the common sign $(-1)^n$. This proves nonzero peak transfer from x to each individual y, not merely a statement about transfer to an entire class.

Equations (16) now give

$$\frac{|E(Y_n)|}{\mathrm{dist}(x, y)^3} = \frac{n(n+1)(2n+1)}{6(2n-2)^3} \longrightarrow \frac{1}{24}.$$

Together with (14), this proves optimality of the universal leading constant, on an infinite sequence of even distances.

6.2 Odd distances and the order of the minimum

Peak transfer at a common time is preserved by Cartesian products. To see this, suppose the nonzero mutual spectral coefficients in two factors are c_θ and b_λ, aligned at τ with phases γ_1, γ_2. The product adjacency matrix is $A \otimes I + I \otimes B$. Before grouping repeated eigenvalue sums, its mutual coefficients are $c_\theta b_\lambda$ at $\theta + \lambda$. For a fixed sum, alignment gives

$$e^{i\tau(\theta+\lambda)} c_\theta b_\lambda = \gamma_1\gamma_2 |c_\theta b_\lambda|.$$

All nonzero summands therefore have the same sign after removing the common phase. Grouping cannot cause cancellation; hence the product also attains its spectral triangle bound.

For an adjacent vertex pair in the Petersen graph, the eigenvalues $3, 1, -2$ have respective mutual projector entries $1/10, 1/6, -4/15$. These follow from $E_3 = J/10$, $E_3 + E_1 + E_{-2} = I$, and $3E_3 + E_1 - 2E_{-2} = A$. At $\tau = \pi$ their phases are $-1, -1, 1$, so this pair has peak transfer. The Cartesian product of Y_n with the Petersen graph consequently has peak transfer over distance $2n - 1$ and has

$$10\frac{n(n+1)(2n+1)}{6} + 15n(n+1) = O(n^3)$$

edges. The even distances are supplied by Y_n itself, and distance one by K_2. This proves $m_A(d) = \Theta(d^3)$ for all positive integer distances. The product construction is used only for the order bound; it does not assert that the constant $1/24$ is attained along odd distances.

Scope and verification. The finite verification script checks the complete unweighted adjacency matrices for $2 \le n \le 6$: their characteristic polynomials, geodesic moments, all mutual projector entries, and exact phase signs. It also checks the optimal weighted-path characteristic polynomials and energy identities for $1 \le d \le 10$. These exact checks supplement the proofs. The result answers the lower-bound request of Open Problem 7.2 and establishes its optimal adjacency growth order. It does not classify all equality graphs, determine the exact minimum for every distance, or prove a sharp Laplacian asymptotic. No claim of a historically unprecedented general trace or Galois principle is made.

7 Attribution and the requested success conditions

The original targets are explicitly numbered in their primary sources: Ulas Conjecture 4.2(1) [1], Clarke Questions 6.6 and 6.7 [2], and Coutinho–Guo–Schmeits Open Problem 7.2 [4]. The proofs use the original Stern indices, the original definition of a minimal invariant subpolytope, the original nonsaturated face-generated group, and the original graph-distance variable.

T81 is a refutation of the whole conjunction through its monotonicity clause. The infinite range $n \ge 13$ is essential: isolated small indices could not refute a claim about all sufficiently large indices. T82 answers the literal yes/no question, but its historical classification requires care. Clarke's already recorded thirteen-dimensional example also becomes minimal under the projection-fiber lemma. Thus the unbounded family and explicit minimality certificate, rather than the raw join counterexample, are the proposed additional content. T83 answers the entire stated three-dimensional existence question. T85 supplies the bounds formally requested in Open Problem 7.2 and the optimal adjacency cubic order; it does not identify every finite-distance extremal graph.

The methodological antecedents are substantial. The Stern expression is already in Ulas. Noncollapse of the second Ehrhart coefficient is prior [3]. Free joins and reflected-square cancellation occur in the Ziegler examples cited by Clarke. Spectral-spacing and trace bounds are in Coutinho [5]; arithmetic mutual-support conditions are in [4]; and the attaining Krawtchouk chain and its fixed-energy constant are prior [6, 7]. The new statements proposed here concern the specific derivative obstruction, the unbounded minimal family, the face-to-parity implication, and the general peak-transfer optimization and its graph consequences. Recombining these tools is not evidence that the underlying general principles are historically new.

Primary-source and follow-up searches through 24 October 2026 located no explicit earlier resolution identical to T81, T83, or the complete T85 bound-and-sharpness statement. This

is a bounded search result, not proof that no such work exists. The overlap for T82 and the known attaining benchmark for T84 are stated rather than hidden. Accordingly, the five theorems are not reported as five successes satisfying every historical condition in the research brief.

Screened branches not promoted to successes. An attempted general-lattice Floquet extension left higher-order Fourier cycles uncontrolled and did not solve the PT-symmetry conjecture. An all-residue Motzkin Hankel analysis did not establish the continued-fraction cycle required by the original conjecture, and its main Christoffel mechanism overlaps Chern–Shi [8]. These branches are omitted from the theorem count. Direct square-completion and fewnomial consequences of Ulas's formulas are likewise not promoted as new methods.

A Exact verification and reproducibility

The accompanying archive contains four Python programs and their recorded JSON outputs. Each uses exact integer, rational, or symbolic arithmetic. The finite checks support the implementation of the formulas; the preceding proofs establish the infinite quantifiers.

Proof component	Executed verification
Stern derivative certificate	Exact rational certificate at $-3/4$, with the uniform error threshold $n = 13$; independent recursive derivatives for $13 \le n \le 100$. The script also records classical auxiliary identities, which are not included in the new-theorem count.
Minimal Ehrhart family	$1 \le m \le 12$: fundamental parallelepiped enumeration, independent exact counts in Q_m and $2Q_m$, and the character, point-count, volume, and degree-two formulas.
Half-lattice boundary	52 half-lattice polygons: both Ehrhart constituents and the exact parity-boundary formula, including a nonintegral-vertex period-collapse example.
Peak-transfer bounds	Full unweighted adjacency matrices for $2 \le n \le 6$: characteristic polynomials, geodesic moments, individual-vertex projectors, and phase signs. Optimal weighted paths for $1 \le d \le 10$.

Separate internal checks rederived the Stern certificate directly from the recursion, the Ehrhart family from its homogeneous lattice, the face-to-pole argument, and the graph bounds from mutual moments and exact gap groups. These are internal mathematical checks, not external peer review. The TEX source is standalone and contains its bibliography; compiling twice with `pdflatex` requires no separate figures or bibliography database. The archive supplies its own run instructions and dependencies.

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